In this assignment, you continue your exploratory data analysis of the stock price data in the two databases "**bsedata1**" and "**nsedata1**" that you have already obtained. You are free to use other such basic statistical tools (like volatility clustering, autocorrelation function, etc), apart from what has been listed below, to gain a better understanding of the behaviour of the stock/index prices, keeping in mind the classical BSM setup.

Now for each of the stocks and for each of the market indices do the following :

1. Consider the daily returns R_i, report descriptive statistics of the data by reporting the values of mean, standard deviation, skewness and kurtosis.

   Do a boxplot of the returns. What are your observations?

   Also generate the quantile-quantile plots. What are your observations?

   Use (you may first learn about these, before using) the Kolmogorov–Smirnov test and the Shapiro–Wilk test to test for the normality assumption of the returns, by reporting the test statistic value and the p-value. What are your observations?

   Report the maximum likelihood estimates of the mean return and the variance of the returns, assuming that the returns follow a Gaussian distribution, along with 95% confidence intervals for the parameters.

2. Will the observations be different if you instead use the log returns ?

3. Repeat the above with weekly and monthly data.

Summarize your observations in your report. Remember that you need (and should) not put all the figures in the report. General representative figures are sufficient, along with figures which show different and interesting behaviours.

Submission Deadline: March 14, 2026, 11:59 PM